

8086 Microprocessor in Maximum Mode

Q. Explain the operation of the 8086 microprocessor in Maximum Mode. Discuss the role of the Bus Controller (8288) and how a multiprocessor environment is managed.

📌 Definition to Remember

The 8086 operates in **Maximum Mode** when the **MN/MX (Pin 33)** pin is tied to Ground (Logic Low). Designed for **multi-processor systems** (e.g., using an 8087 coprocessor), the 8086 stops generating control signals and instead sends status codes to an external **8288 Bus Controller**.

1. Maximum Mode Configuration

In this mode, pins that previously generated control signals (like **ALE**, **WR**) take on new functions related to bus arbitration and multiprocessor coordination.

* The 8086 relies on the **Intel 8288 Bus Controller** to generate actual memory and I/O signals, freeing the CPU to manage shared resources.

2. Reassigned Maximum Mode Signals

- **S2', S1', S0' (Status Lines):** The 8086 outputs its current cycle status on these 3 pins. The 8288 reads them to generate command signals.
- **RQ'/GT0' & RQ'/GT1' (Request/Grant):** Replaces HOLD/HLDA. Used by coprocessors to request control of the local bus from the 8086, and for the 8086 to grant it.
- **LOCK':** When the 8086 executes an instruction with a **LOCK** prefix, it asserts this pin. It prevents other processors from taking the bus, ensuring safe updates to shared memory.
- **QS1, QS0 (Queue Status):** Broadcasts the status of the 8086's internal instruction queue to coprocessors.

3. Role of the 8288 Bus Controller

The 8288 decodes the **S2'-S0'** signals to generate system controls:

* **000:** Interrupt Acknowledge (INTA')

* **001:** Read I/O Port (IORC')

* **010:** Write I/O Port (IOWC')

* **101:** Read Memory (MRDC')

* **110:** Write Memory (MWTC')

(The 8288 also generates **ALE**, **DEN**, and **DT/R'** for the system).

4. Managing the Multiprocessor Environment

- **Bus Arbitration:** To prevent data corruption, only one processor can use the bus at a time. A coprocessor asserts **RQ'**. The 8086 finishes its cycle, floats its buses (High-Z), and asserts **GT'** to hand over control.
- **Shared Memory:** The **LOCK'** signal is crucial. It creates an "atomic" operation, preventing other CPUs from reading a memory location while the 8086 is in the middle of updating it.

★ **Must-Write Points (for 10 marks)**

1. **Maximum Mode** is activated by tying the **MN/MX (Pin 33)' pin to Ground (Low)**.
2. Designed for **multi-processor** environments (e.g., CPU + 8087 Math Coprocessor).
3. The 8086 stops generating control signals; relies on the **8288 Bus Controller**.
4. 8086 outputs status on **S2, S1, S0**, which the 8288 decodes to generate Memory/IO Read/Write signals.
5. HOLD/HLDA are replaced by bidirectional **RQ'/GT' (Request/Grant)** for bus arbitration.
6. **LOCK' pin**: Asserts exclusive control over the bus to protect shared memory.
7. **QS0/QS1**: Broadcast instruction queue status to coprocessors.

⚡ **Quick Recall**

Maximum Mode → MN/MX (Pin 33)' = GND → Multi-processor → 8288 Bus Controller does the work → CPU sends S0, S1, S2 → Uses RQ/GT for bus request → LOCK to protect shared memory